

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
```

```
def predict(X, weights, lambd=1):
    net = np.dot(X, weights)
    return (1 / (1 + np.exp(-lambd * net)))

def calculate_dw(X, y, y_pred):
    return ((y_pred - y) * (y_pred) * (1 - y_pred) * X)

def calculate_error(yi, y_pred):
    return np.square(y_pred - yi)
```

```
def train_vanilla(X, y, weights, epochs=500, lr=0.01, epsilon=1e-8):
    vel = 0
    for epoch in range(epochs):
        error = 0
        dw = 0
        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, weights)
            dw += calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)

        vel = vel + (dw**2)
        weights -= ((lr * dw) / (np.sqrt(vel + epsilon)))
        error /= (2 * len(X))

        if (epoch+1) % 50 == 0:
            print(f'Weights after epoch {epoch+1} : ',weights)
            print(f'Error after epoch {epoch+1} : ',error)
```

```
def train_stochastic(X, y, weights, epochs=500, lr=0.01, epsilon=1e-8):
    vel = 0
    for epoch in range(epochs):
        error = 0
        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, weights)
            dw = calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)
            vel = vel + (dw**2)
            weights -= ((lr * dw) / (np.sqrt(vel + epsilon)))

        error /= (2 * len(X))

        if (epoch+1) % 50 == 0:
            print(f'Weights after epoch {epoch+1} : ',weights)
            print(f'Error after epoch {epoch+1} : ',error)
```

```
def train_min_batch(X, y, weights, epochs=500, lr=0.01, epsilon=1e-8, bs=32):
    vel = 0
    for epoch in range(epochs):
        error = 0
        i = 0
        dw = 0
        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, weights)
            dw += calculate_dw(Xi, yi, y_pred)
            i += 1
            error += calculate_error(yi, y_pred)

        if i % bs == 0 or i == len(X):
            vel = vel + (dw**2)
            weights -= ((lr * dw) / (np.sqrt(vel + epsilon)))
            dw = 0

        error /= (2 * len(X))

        if (epoch+1) % 50 == 0:
            print(f'Weights after epoch {epoch+1} : ',weights)
            print(f'Error after epoch {epoch+1} : ',error)
```

```
df = pd.read_csv('bank_note.csv')
df.insert(4, 'x0', 1)
```

```
X = df[['variance','skewness','curtosis','entropy','x0']].values
y = df['class']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=13, test_size=0.2)
weights = input('Enter 4 weights and 1 bias : ').split()
weights = np.array([float(weight) for weight in weights], dtype='longdouble')
```

```
print('Adagrad (Vanilla)')
train_vanilla(X_train,y_train,weights.copy())
```

```
➡ Adagrad (Vanilla)
Weights after epoch 50 : [ 0.37125579 -0.30838689  0.65753559 -0.46097721  0.16952008]
Error after epoch 50 : 0.1946341730544026
Weights after epoch 100 : [ 0.3120671 -0.34736975  0.58641092 -0.53477082  0.25766423]
Error after epoch 100 : 0.177002668515983
Weights after epoch 150 : [ 0.26777458 -0.37490175  0.53146222 -0.58634531  0.31704715]
Error after epoch 150 : 0.16372353833392655
Weights after epoch 200 : [ 0.2315893 -0.39743771  0.48585268 -0.6254709  0.36178188]
Error after epoch 200 : 0.153139094685064
Weights after epoch 250 : [ 0.20068647 -0.41692467  0.4465855 -0.65643305  0.39753448]
Error after epoch 250 : 0.14444958067308356
Weights after epoch 300 : [ 0.17355687 -0.43409854  0.41192517 -0.68160111  0.42725289]
Error after epoch 300 : 0.13715609150323957
Weights after epoch 350 : [ 0.14928049 -0.44932643  0.38075313 -0.70243242  0.45264718]
Error after epoch 350 : 0.1309207743162474
Weights after epoch 400 : [ 0.12724533 -0.46285154  0.35230795 -0.71988276  0.47476727]
Error after epoch 400 : 0.12550543682790888
Weights after epoch 450 : [ 0.10701843 -0.47486452  0.32605111 -0.73460918  0.49428376]
Error after epoch 450 : 0.12073733494995528
Weights after epoch 500 : [ 0.08827947 -0.48552444  0.30158972 -0.7470809  0.51163933]
Error after epoch 500 : 0.11648826328538676
```

```
print('Adagrad (Stochastic)')
train_stochastic(X_train,y_train,weights.copy())
```

```
➡ Adagrad (Stochastic)
Weights after epoch 50 : [-0.7522853 -0.42558678 -0.42320246 -0.22794774  0.5041294 ]
Error after epoch 50 : 0.016979124903761002
Weights after epoch 100 : [-0.89326723 -0.48221969 -0.52334002 -0.15692226  0.77670195]
Error after epoch 100 : 0.01243945084095838
Weights after epoch 150 : [-0.97274577 -0.52046054 -0.58476738 -0.11925709  0.9517575 ]
Error after epoch 150 : 0.01050620157066231
Weights after epoch 200 : [-1.02972711 -0.54994677 -0.62966331 -0.09477389  1.07853071]
Error after epoch 200 : 0.00940011054135062
Weights after epoch 250 : [-1.07497758 -0.57412455 -0.66527309 -0.07729239  1.17672985]
Error after epoch 250 : 0.00867333281926056
Weights after epoch 300 : [-1.11291578 -0.59470341 -0.69490321 -0.06408599  1.25617029]
Error after epoch 300 : 0.008154378775331586
Weights after epoch 350 : [-1.14579082 -0.61266864 -0.72034581 -0.05372172  1.32242719]
Error after epoch 350 : 0.007762463034424446
Weights after epoch 400 : [-1.17491504 -0.62864304 -0.74268373 -0.04536077  1.37895802]
Error after epoch 400 : 0.007454275792636227
Weights after epoch 450 : [-1.20112876 -0.6430466 -0.76262265 -0.0384739  1.42804699]
Error after epoch 450 : 0.007204408704021717
Weights after epoch 500 : [-1.22500695 -0.65617654 -0.78064893 -0.03270802  1.47127731]
Error after epoch 500 : 0.006996924682271778
```

```
print('Adagrad (Mini Batch)')
train_min_batch(X_train,y_train,weights.copy())
```

```
➡ Adagrad (Mini Batch)
Weights after epoch 50 : [-0.2076353 -0.41253244 -0.06397104 -0.61871423  0.26750986]
Error after epoch 50 : 0.061492952189977766
Weights after epoch 100 : [-0.41797584 -0.3978772 -0.28350786 -0.43557702  0.32120382]
Error after epoch 100 : 0.031569190930990476
Weights after epoch 150 : [-0.53659205 -0.41534542 -0.364503 -0.32382931  0.47048517]
Error after epoch 150 : 0.022653973120027985
Weights after epoch 200 : [-0.61725333 -0.43239098 -0.41186312 -0.25828953  0.60365222]
Error after epoch 200 : 0.018438925766740093
Weights after epoch 250 : [-0.67781422 -0.44754751 -0.44742683 -0.21417796  0.71229964]
Error after epoch 250 : 0.015940292850899377
Weights after epoch 300 : [-0.72624402 -0.46140298 -0.47663618 -0.18201622  0.80238334]
Error after epoch 300 : 0.014274154380791331
Weights after epoch 350 : [-0.76665919 -0.47425929 -0.50169217 -0.15735871  0.8787882 ]
Error after epoch 350 : 0.013076344482321
Weights after epoch 400 : [-0.80141551 -0.48628381 -0.52375673 -0.13777835  0.94484933]
Error after epoch 400 : 0.01216903892103128
Weights after epoch 450 : [-0.8319722 -0.49758838 -0.54353838 -0.12181549  1.00285838]
Error after epoch 450 : 0.011454965367686204
Weights after epoch 500 : [-0.85929057 -0.50825752 -0.56150849 -0.10853173  1.05443898]
Error after epoch 500 : 0.010876328368886553
```